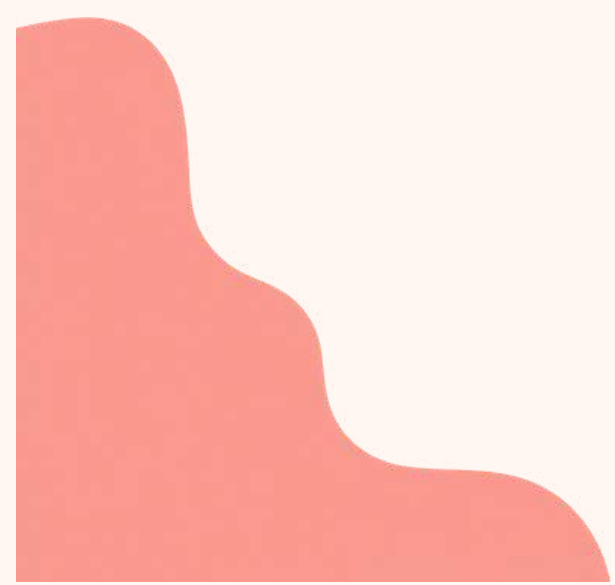




MASTER THE ORDER: STACKS & QUEUES IN ACTION!

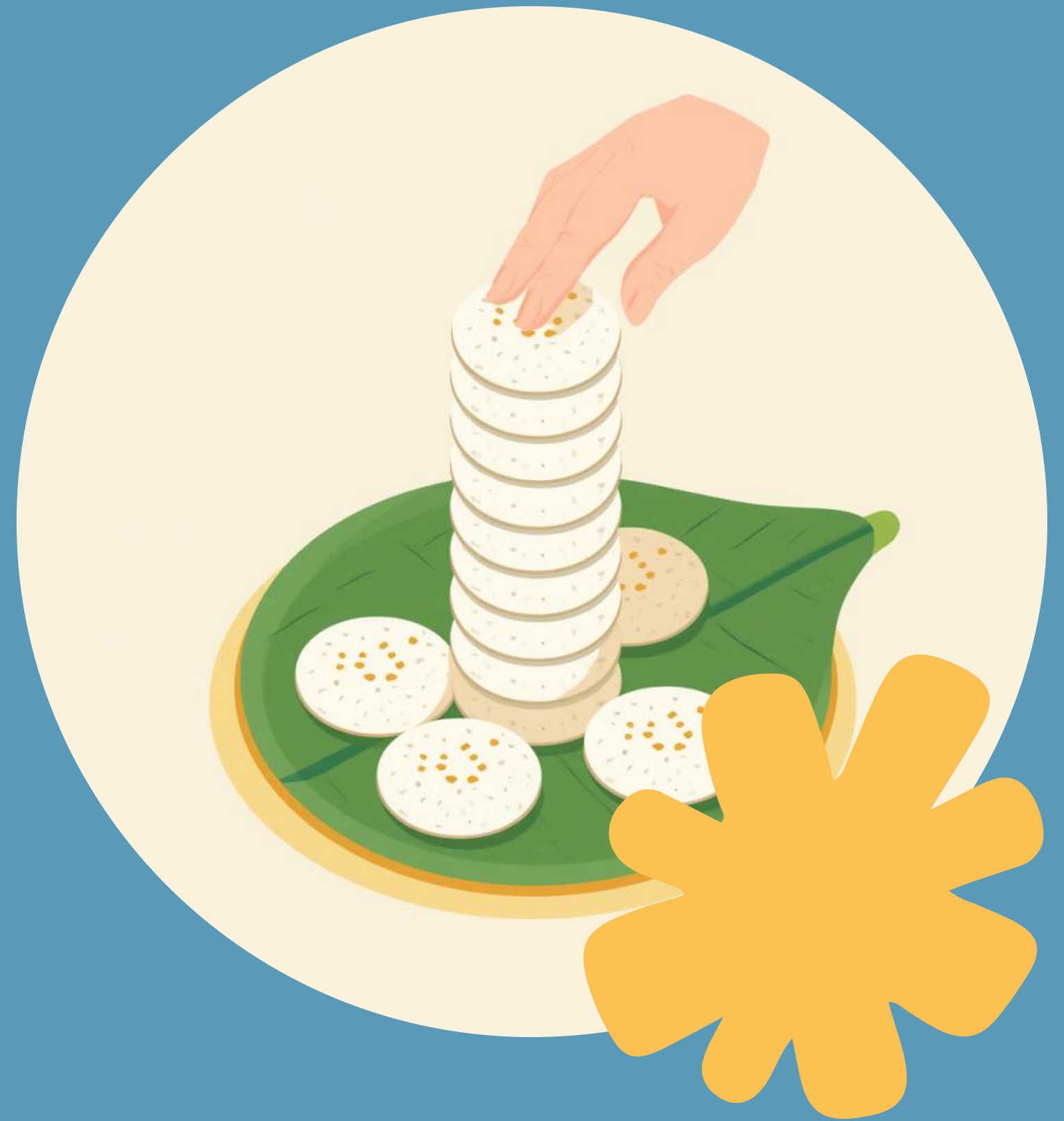


WHAT COMES
OUT LAST GOES
IN FIRST? 🥞
WHAT COMES
OUT FIRST GOES
IN FIRST? 🧒

WHAT IS A STACK?

STACK – LAST IN, FIRST OUT (LIFO)

Think of a stack of hot idlis! You always take the one on top first. The last idli placed on top is the first one you eat. That's exactly how a stack works in computers — the last item added is the first one removed!





STACK OPERATIONS: PUSH, POP & PEEK



PUSH

Add a new idli on top of the stack. The newest item sits at the top!



PEEK

Look at the top idli without removing it. Just a quick peek!

POP

Remove the top idli from the stack. Last one in is the first one out!



UNDO WITH A STACK!

Every time you press Undo in a game or app, a stack is at work!

Each move you make gets pushed onto the stack.
When you undo, the last move is popped off first.
Last action in = First action undone!

Stack rule: Last In, First Out (LIFO) —
just like undoing your moves!



WHAT IS A QUEUE?

QUEUE – FIRST IN, FIRST OUT (FIFO)

Think of kids standing in a neat line at the temple for prasad — the first one to join the line is the first one to get prasad! That's exactly how a Queue works: the first item added is the first one to come out.



QUEUE OPERATIONS

Every queue has four key moves! Learn how people join, leave, and hold their spot in line.



World Area



PRINTER QUEUE!

Printer jobs wait in a queue — First In, First Out!
When you send documents to print, each job joins the end of the queue. The first document sent is the first one printed. Just like waiting in line — no cutting!
This is exactly how FIFO works in real life.

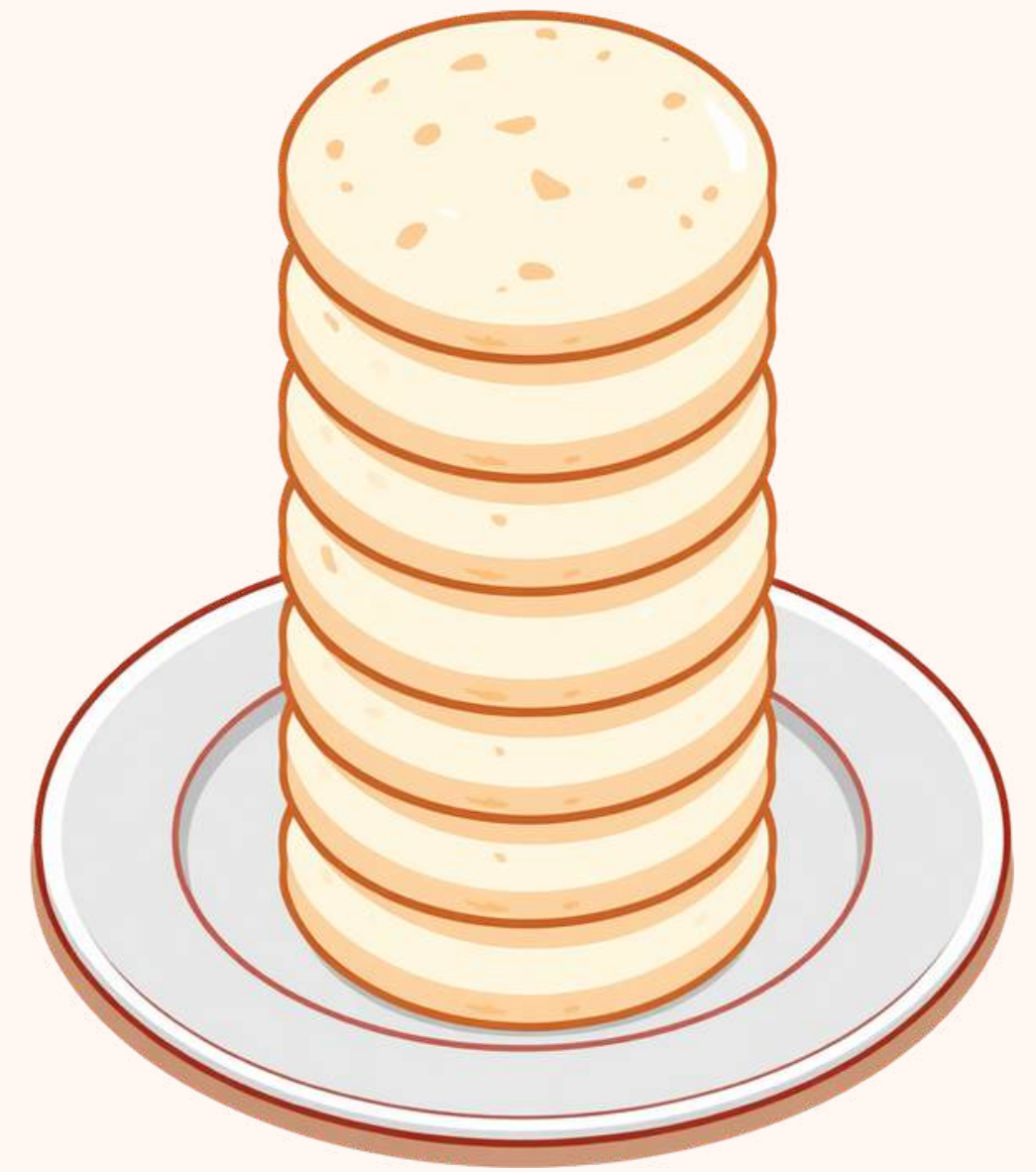
Queue = First In, First Out (FIFO). The printer always prints in order!



STACK (LIFO)

Last In, First Out

- Add on top → Push
- Remove top → Pop
- Like a stack of hot idlis — you take the top one first!
- Example: Undo button in games



QUEUE (FIFO)

First In, First Out

- Join at end → Enqueue
- Leave from front → Dequeue
- Like a queue at the temple for prasad!
- Example: Printer jobs, ticket lines



DECIDE: STACK OR QUEUE?



 Scenario 1:

You press Undo in a drawing app

 Scenario 2:

Waiting in line for a metro ticket

 Scenario 3:

Bangles stacked on a bangle holder

Stack or Queue?

You decide!



CHALLENGE ANSWERS & WHY!



SCENARIO 1:
UNDO BUTTON  STACK

The last action you did is undone first — that's LIFO! Just like taking the top idli off the stack.

SCENARIO 2:
TICKET LINE  QUEUE

The first person to join the line gets served first — that's FIFO! Just like a metro ticket queue.

SCENARIO 3:
BANGLES STACKED  STACK

The last bangle put on your wrist is removed first — that's LIFO! Stack all the way!

SUMMARY & RECAP



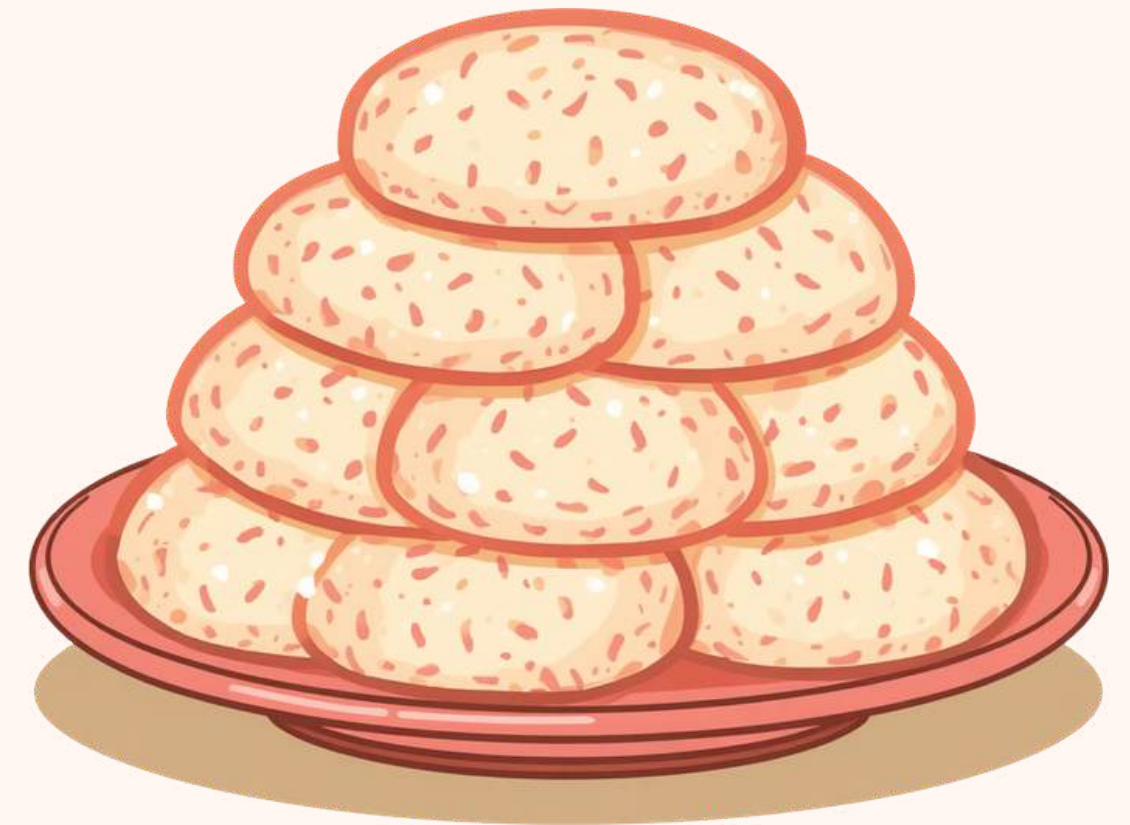
Stack — LIFO

Push: Add to top

Pop: Remove from top

Peek: See the top

Example: Stack of idlis, Undo button



STACK VS QUEUE



Queue — FIFO

Enqueue: Join the end

Dequeue: Leave the front

Front & Rear: First & last in line

Example: Prasad line, Printer jobs





QUICK QUIZ: TEST YOUR KNOWLEDGE!

QUESTION 1

Which structure lets you undo
your last move?

A) Stack B) Queue C) Array

QUESTION 2

Which is like waiting in line for prasad
at the temple?

A) Stack B) Queue C) Both

QUESTION 3

What does "pop" do in a stack?

A) Add to top B) Remove top
C) See top





GREAT JOB! 🎉

